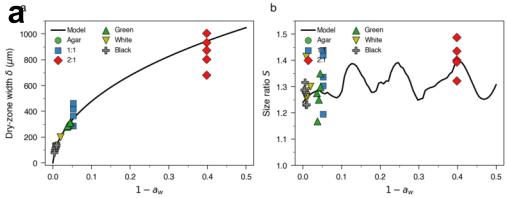
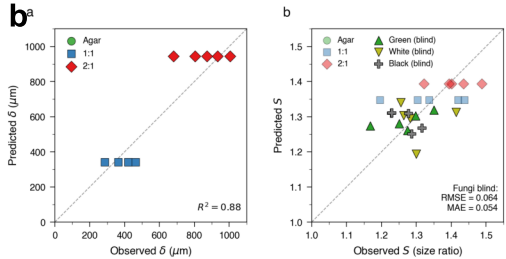
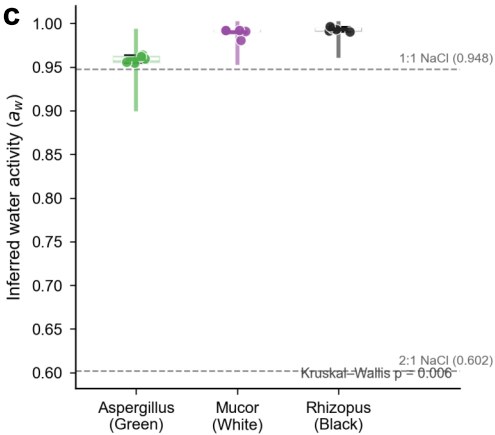
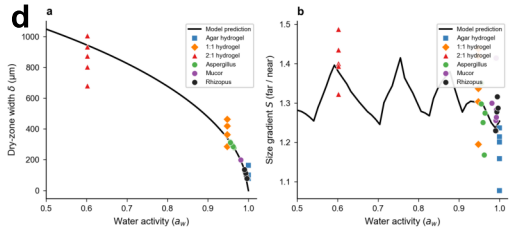

a
Model
Agar
1:1
2:1
Green
White
Black
Dry-zone width δ (μm)
1 − a_w
b
Size ratio S
1 − a_w


b
a
Agar
1:1
2:1
Predicted δ (μm)
Observed δ (μm)
R² = 0.88
b
Agar
1:1
2:1
Green (blind)
White (blind)
Black (blind)
Predicted S
Observed S (size ratio)
Fungi blind:
RMSE = 0.064
MAE = 0.054


c
Inferred water activity (a_w)
1:1 NaCl (0.948)
2:1 NaCl (0.602)
Kruskal–Wallis p = 0.006
Aspergillus (Green)
Mucor (White)
Rhizopus (Black)


d
a
Model prediction
Agar hydrogel
1:1 hydrogel
2:1 hydrogel
Aspergillus
Mucor
Rhizopus
Dry-zone width δ (μm)
Water activity (a_w)
b
Size gradient S (far / near)
Water activity (a_w)